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Reliable and non-destructive Raman analysis to determine the urea concentration in a cream formulation

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We present a reliable and nondestructive analytical method for the determination of urea concentration in a pharmaceutical cream formulation using Raman spectroscopy. A pharmaceutical cream is a highly viscous emulsion; therefore, its composition and physical mixing could be inhomogeneous on a microscopic scale. The local environment around the urea could vary, which could influence the molecular vibrations of the urea molecule. As expected, when Raman spectra were collected by focusing the laser onto a tiny area (∼2–3 μm), the position of the urea band at 1003 cm^{-1} varied as a result of the microscopic inhomogeneity within the sample. Therefore, acquisition of Raman spectra representative of the entire sample rather than a localized portion of it is very important for the analysis of pharmaceutical creams. Based on the preliminary Raman mapping results of a urea cream, a sample area of at least 750 × 750 μm should be covered for reliable quantitative analysis. In this study, we used a wide-area illumination scheme capable of covering a sample area of 28.3 mm^2 for Raman spectral collection in order to ensure a reliable representative sample. In addition, to simplify the measurements, Raman spectra of urea creams in plastic bottles were directly collected without further sampling, and partial least squares regression was used for quantitative analysis. The urea concentrations were accurately determined despite the spectral collection being performed through plastic bottles. **Copyright © 2010 John Wiley & Sons, Ltd.**

Keywords: urea cream; cream formulation; sample representation; emulsion; wide area illumination

Introduction

In this paper, we describe the development of a reliable analytical method for quantitative determination of the active pharmaceutical ingredient (API) concentration in a pharmaceutical cream using Raman spectroscopy. A pharmaceutical cream is a highly viscous emulsion, in which the oil and aqueous phases are simultaneously homogenized. Its composition and physical appearance are homogeneous on a macroscale; however, they could be inhomogeneous on a microscopic scale. The variation in microstructure and the size of the emulsions have been previously studied utilizing a micro-confocal Raman system.[1,2] Variation in the microenvironment could lead to uncertainty in quantitative Raman analysis, as the conventional Raman scheme has a small laser illumination area (80–200 μm). In such a small sampling space, the Raman spectrum would not be representative of the overall composition of the cream sample. In the case of a less viscous suspension of medroxyprogesterone acetate, the sample was stirred during spectral collection to improve sample representation[3]; however, stirring highly viscous creams during spectral collection is impractical. Raman spectroscopy has also been used to monitor the manufacture of topical gels and emulsions[4,5]; however, these samples were much less viscous than a typical pharmaceutical cream.

To improve the sample representation and reduce the subsampling problem, spatially offset Raman spectroscopy (SORS) and transmission Raman spectroscopy have been developed.[6] SORS involves the collection of Raman spectra from regions spatially offset on the sample surface from the laser beam interaction point, thereby leading to a wider spread of photons interacting with deeper layers emerging out of the sample surface. Transmission Raman measurement is based on the collection of Raman signal at the opposition side of the laser illumination, so that Raman spectra representing bulk content of samples can be acquired. In addition, a wide-area illumination (WAI) scheme, in which a laser simultaneously illuminates a large sample area, e.g. 28.3 mm^2, has been used to enhance the sample representation for the analysis of diverse pharmaceutical samples.[7–11]

In this paper, we describe attempts to nondestructively determine the urea concentration in a cream formulation using Raman spectroscopy. Urea-based creams are topical pharmaceuticals used for the treatment of many skin diseases, such as dermatitis,[12] eczema,[13] xerosis,[14] hyperkeratotic type tinea pedis[15] and recalcitrant verruca vulgaris.[16] The British pharmacopoeial procedure for quantitatively analyzing a urea cream is a colorimetric method,[17] which requires a great deal of sample derivatization; another method, hydrophilic interaction

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chromatography, is likewise time consuming for sample and mobile-phase preparation.[18,19] Therefore, a fast and reliable analytical method, such as Raman spectroscopy, would be of great interest.

The microscopic homogeneity of a urea cream was initially investigated using Raman mapping (sample coverage area: 100 × 150 μm) with the help of a microscope. This was to examine the possible variation in the urea band on a microscopic scale and to address the requirement of accurate sample representation of a urea cream for reliable quantitative analysis. Raman images obtained by mapping six different areas of the same cream sample were compared, and the possible variations in the urea band, such as shifting of the band position, due to an inhomogeneous chemical environment around the urea was examined.

To effectively ensure suitable sample representation and accurate quantitative analysis of the urea concentration, a WAI scheme, in which a laser illuminates a 6-mm-diameter circular spot (total coverage area: 28.3 mm^2), was used to collect Raman spectra of cream samples. To enhance the simplicity of the analysis, the urea creams were directly measured in their plastic containers, eliminating the manual sample transfer into a spectroscopic cell. Since the urea cream is highly viscous, its physical transfer into a cuvette for spectral collection is somewhat cumbersome, and washing the cuvette after the measurement is also tedious. To assess the possible decrease in accuracy in the determination of urea concentration due to performing the measurement in a round plastic container, the accuracies of the measurements for the same samples in a plastic container and a quartz cuvette were compared.

Experimental

Samples

The preparation of the oil phase was done by mixing stearic acid, white vaseline, isopropyl myristate, propylene glycol, white wax, triethanolamine and concentrated paraben (10 : 1). The concentrated paraben consisted of 10% w/w methyl paraben and 1% w/w propyl paraben, in propylene glycol. The aqueous phase was prepared by dissolving appropriate amounts of urea in purified water (55.5% w/w). The oil and aqueous phases were heated separately to 70–75 °C, and then gradually mixed together with continuous stirring until the mixture congealed. For each sample, 6.9 g of the oil phase was homogenized with the aqueous phase to yield 20 g of the final urea cream. Fifteen different concentrations of urea cream ranging from 7.2 to 12.8 wt% were prepared in 0.4 wt% increments. The urea creams were transferred into low-density polyethylene (LDPE) bottles (diameter: 22 mm, height: 46 mm).

Raman spectral collection and data processing

A WAI Raman scheme (Kaiser Optical Systems, Ann Arbor, MI, USA) that illuminates a 6-mm-diameter spot with the laser (total illumination area: 28.3 mm^2) was used to collect Raman spectra of the urea creams. A detailed explanation of the WAI scheme can be found in our previous publication.[7]

Initially, Raman mapping was performed over a sample area of 100 × 150 μm. The point-to-point distances in both the *x* and *y* directions were 5 μm, thereby resulting in a total of 600 data points (20 × 30 points). Raman spectra were collected using a dispersive Raman microscope (Kaiser Optical Systems) equipped with a diode laser (λ_{ex} = 785 nm), a thermoelectrically cooled (−40 °C) CCD detector and a holographic grating. The urea cream sample was positioned on a microscope stage, and the laser beam was focused using an objective lens (10 × /0.25 numerical aperture) to collect Raman spectra with a resolution of 4 cm^{-1}.

Raman spectra of the urea creams were collected in two different ways using the WAI scheme. Initially, the urea creams were directly measured in their plastic bottles. A plastic bottle containing the urea cream was located at the focal point of the WAI scheme for spectral collection, and spectra were collected in triplicate for each sample. During the triplicate measurements, three different spots on the plastic bottle (0°, 120° and 240°) were illuminated, and the corresponding spectra were collected at each position. To examine the effect of the plastic container on quantitative analysis, we also collected in triplicate Raman spectra of the same samples manually transferred into a quartz cuvette (10 × 10 mm). The measurements with the quartz cuvette were used to determine whether accuracy had been compromised by data acquisition through the plastic bottles. In both cases, Raman spectra were collected using a 785-nm laser with an exposure time of 10 s (total 60 s for acquisition, resolution: 4 cm^{-1}).

All calculations, including baseline correction, intensity normalization and partial least squares (PLS) regression, were performed in Matlab version 7.0 (Math-Works Inc., MA, USA).

Results and Discussion

Raman spectral features of the urea cream

Figure 1(a) shows the normalized Raman spectra for three different concentrations (7.2, 10.0 and 12.8 wt%) of urea creams, collected by using the WAI scheme. The Raman spectra in the figure were collected using a quartz cuvette. The baselines of the raw Raman spectra were initially corrected at two different wavenumbers (1545 and 755 cm^{-1}), and the peak areas under the 1545–755 cm^{-1} range were calculated. Each baseline-corrected spectrum was divided by the corresponding peak area. Triplicate spectra of each sample are shown in this figure, and the urea band is enlarged in the 1030–980 cm^{-1} range. As the urea cream was composed of several different components as stated in the experimental section, the spectral feature of urea cream is the sum of the spectral features of each component including urea, weighted by the concentrations of each component. As clearly shown, the strongest urea band at 1003 cm^{-1} (N–C–N stretching) clearly increases with concentration, whereas the intensities of the other bands around 1450 and 1295 cm^{-1}, mainly from the organic cream phase, decrease accordingly. It is known that weak urea bands are also present at 1648, 1540 and 547 cm^{-1}[20]; however, these are not clearly seen because of their low intensity as well as overlapping with the bands of other cream components. Obviously, the most intense 1003 cm^{-1} urea band should be the most valuable for quantification. Although the urea band at 1003 cm^{-1} is fairly distinct in terms of shape, it still overlaps with minor bands from other components.

Examination of the microscopic homogeneity of urea cream

Normally, Raman spectra are collected by illuminating a small area of the sample by a focused laser beam and collecting the scattered radiation, allowing for the collection of only local information, especially when the sample is not homogeneous on a microscopic scale. A pharmaceutical cream such as a urea-based cream is

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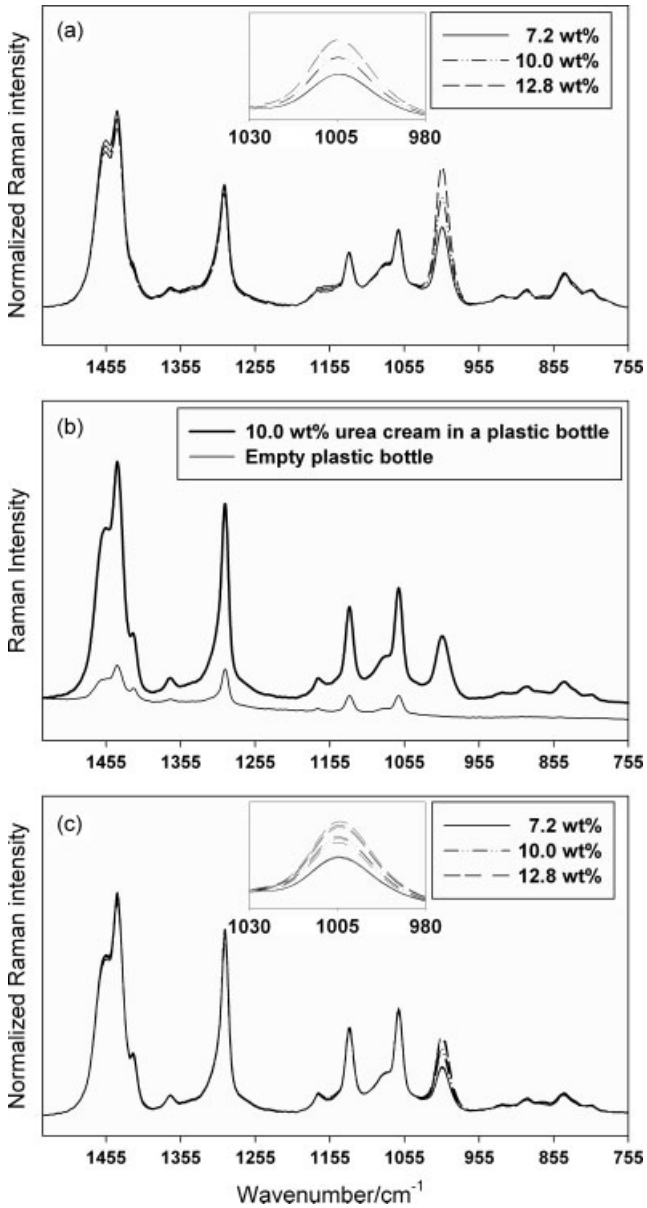

Figure 1. Normalized Raman spectra of three different concentrations (7.2, 10.0 and 12.8 wt%) of urea creams collected (a) in a quartz cell and (c) directly through plastic bottles. In both cases, triplicate spectra of each sample are shown and the urea band in the 1030–980 cm^{-1} range is enlarged. (b) Uncorrected (raw) Raman spectra of a plastic bottle and a 10.0 wt% urea cream-filled plastic bottle.

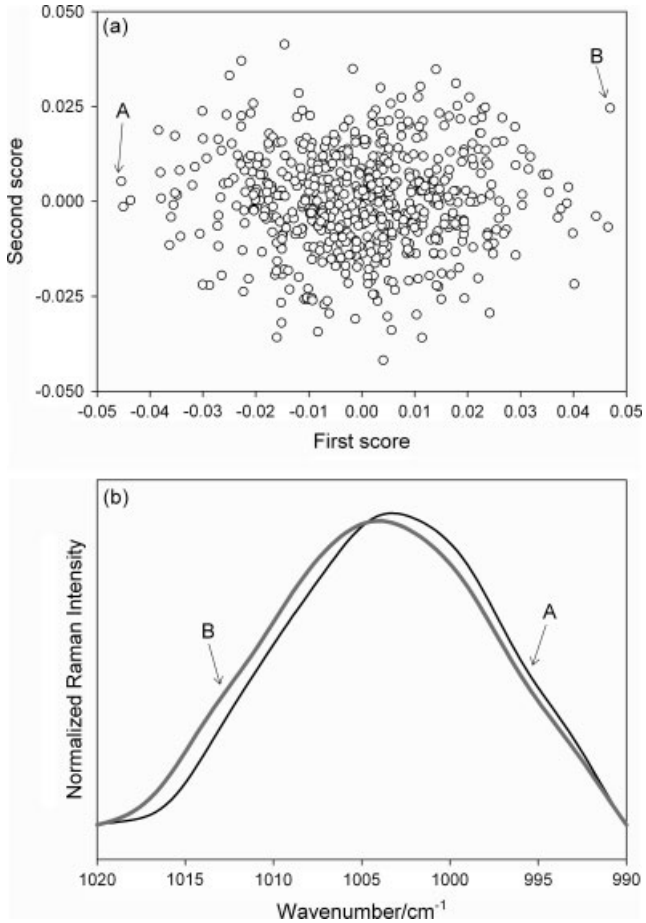

Figure 2. (a) Two-dimensional score scatter plot (the first *vs* second scores) generated from 600 spectra collected for Raman mapping over an area of 100 × 150 μm. Two scores, A and B, correspond to the lowest and highest first score values, respectively. (b) The corresponding Raman spectra of A and B are shown.

a highly viscous emulsion incorporating aqueous and organic phases; therefore, its local physical presentation relative to the size of the laser illumination may not be homogeneous. Additionally, the chemical environment surrounding the API (urea, in this study) could vary, leading to a variation in the intermolecular interactions of the API with the surrounding chemical matrix.

To examine the possible local microscopic inhomogeneity in the cream matrix, we obtained a Raman image of the 10.0 wt% urea cream by mapping using a microscope. The sample was transferred into a metal container, and the top of the container was covered with a 100-μm slide glass to prevent water evaporation during the spectral collection. The mapping was performed over a flat sample area of 100 × 150 μm. The point-to-point distance was 5 μm, thereby resulting in a total of 600 data points (20 × 30 points). The size of the laser spot was approximately 2–3 μm. The same mapping procedure was repeated at six different areas of the cream.

To generate Raman images from each mapping, we used the first score generated from principal component analysis (PCA).[21,22] As previously mentioned, a total of 600 spectra collected at each map were used for the PCA. Before performing PCA, the urea band centered at 1003 cm^{-1} was normalized. The baseline of each spectrum was corrected to zero at 1020 and 990 cm^{-1}, and then the peak area in the 1020–990 cm^{-1} range was calculated. Each baseline-corrected spectrum was divided by the corresponding peak area. The two-dimensional score scatter plot (the first *vs* second scores) acquired from one Raman mapping (total of 600 spectra) is presented in Fig. 2(a). Two scores designated A and B correspond to the lowest and highest first score values, respectively, and the corresponding Raman spectra are shown in Fig. 2(b). As shown in the figure, the positions of the urea band are different in both cases. As the urea cream can be regarded as microscopically bi-phasic or locally inhomogeneous, the chemical surroundings of the urea are not always consistent, which could induce the observed shift of the urea band in Fig. 2(b).

Figure 3 shows six different Raman images (100 × 150 μm) acquired at different areas of a 10.0 wt% urea cream sample. The

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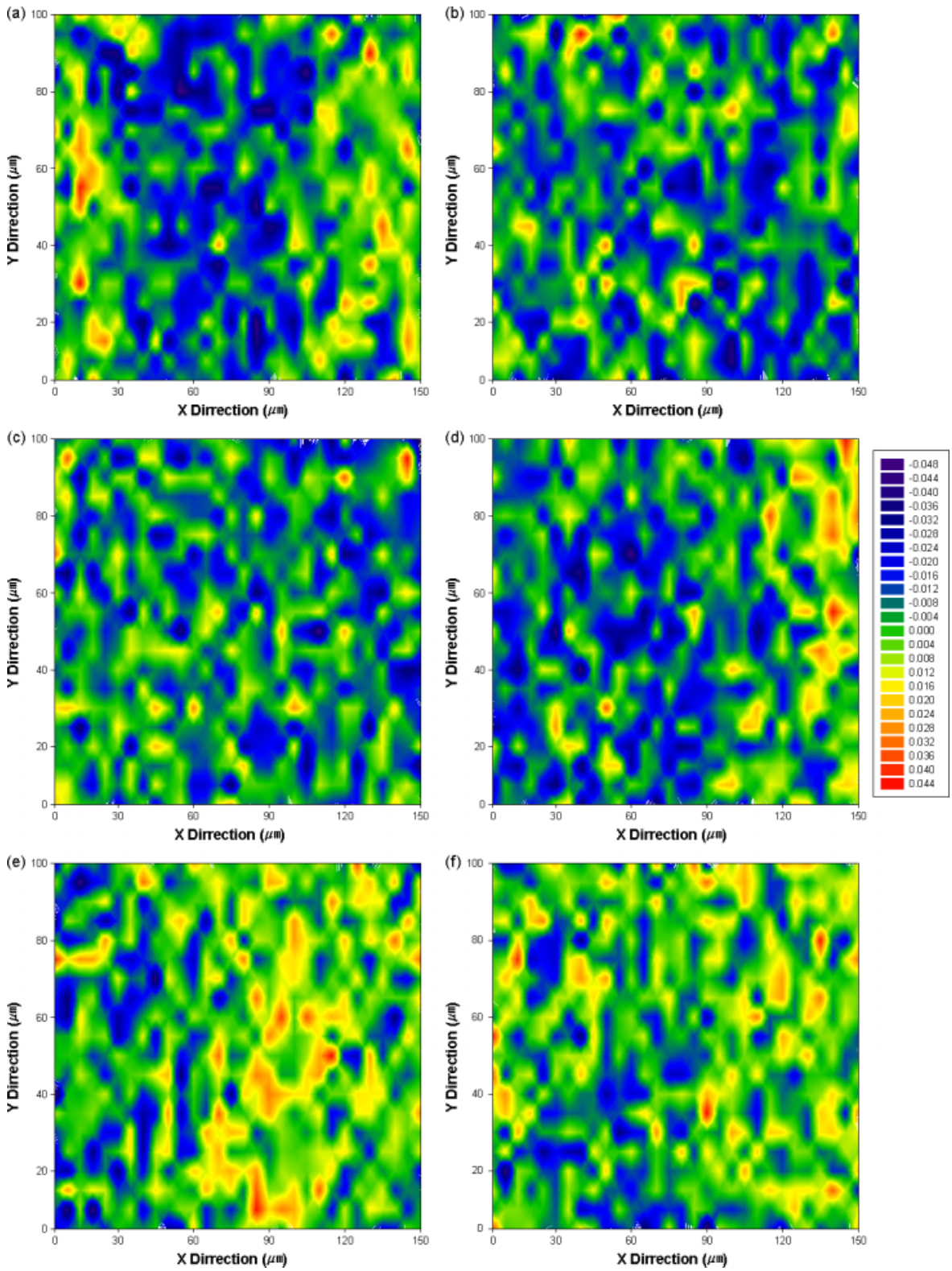

Figure 3. Six different Raman images(a, b, c, d, e and f; image area: 100 × 150 μm) acquired at different areas of a 10.0 wt% urea cream. The images were generated using the first scores of the collected spectra in each mapping. The red color indicates that the urea band is positioned at the highest wavenumber in a given map, while purple color indicates the band position located at the lowest wavenumber.

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images were generated using the first scores of the collected spectra. The red color indicates that the urea band is positioned at the highest wavenumber, whereas purple indicates that the band position is located at the lowest wavenumber. As shown in the images, the position of the urea band varies even with a narrow field of view (100 × 150 μm). In addition, the variation in the band position is almost random, without any regular pattern. In the case of image (a), the purple and blue colors are relatively more dominant; the red and yellow colors are relatively more prominent in image (e).

Keuleers *et al*. observed that the urea band shifted to a lower wavenumber when the sample was changed from crystalline urea to an aqueous urea solution.[23] As the force constant of N–C–N symmetric stretching for urea in an aqueous solution was relatively weak compared to that in crystalline form, the urea band in water shifted to a lower wavenumber. Based on this observation, when the surrounding chemical matrix provides relatively weak intermolecular interactions with the N–C–N symmetric stretching, the corresponding band would shift to a slightly lower wavenumber. As the chemical environment of the urea molecule in the cream formulation is not always consistent, the resulting interaction of urea with its surroundings is not uniform, which could explain the variation in the urea band seen in Figs 2. It is practically difficult to elucidate individual urea interactions with surrounding molecules because the cream matrix is highly complex in chemical composition and physical presentation. However, it is reasonable to presume that a more hydrophilic environment provides relatively strong intermolecular interactions with the urea, which then shift the urea band to a higher wavenumber, such as the samples around score B in Fig. 2; samples around score A are exposed to a more hydrophobic environment, resulting in weaker intermolecular interactions with the urea, which shift the urea band to a lower wavenumber.

We examined the average spectrum obtained by averaging 600 spectra collected in each mapping. Figure 4(a) shows six different average spectra from six mappings in the 1020–990 cm^{-1} range. For a more detailed examination of the spectral features, the 1005–1001 cm^{-1} range is enlarged inside the figure. We also examined another average spectrum by averaging all the spectra (total of 3600 spectra) from six mappings, and the resulting spectrum (the gray line on top) is shown in Fig. 4(a), slightly offset for clarity. This average spectrum is equivalent to a mapping over a 600 × 900 μm area. The vertical dotted line indicates 1003 cm^{-1}.

As the spectra in each mapping are averaged, the degree of spectral variation, such as a shift of the urea band, among the six different averaged spectra should be relatively small compared to that in each individual mapping (refer to Fig. 2). As shown in Fig. 4(a), the positions of the urea bands are quite close to each other; however, they are not consistent. The slight variation in urea band position could also be expected from the color distribution of the six Raman images in Fig. 3. This result clearly demonstrates that a sample coverage area of 100 × 150 μm would not be sufficient to represent the entire urea cream when Raman spectroscopy is used. The laser spot size of a conventional Raman scheme without a microscope is approximately 80–200 μm; therefore, a strategy for covering a wider sample area would be required to ensure an accurate sample representation and a reliable quantitative analysis.

In this study, to easily improve the sample representation, a WAI scheme capable of illuminating a laser onto a total coverage area of 28.3 mm^2 was used to collect Raman spectra of cream samples.[7] To examine the sample presentation of the WAI scheme, 10.0 wt% urea cream was spread onto an aluminum plate over a 4 × 6 cm area with a thickness of approximately 1.5 mm. Raman spectra were collected using the WAI scheme by randomly illuminating 10 different locations across the sample. Figure 4(b) shows the normalized spectra (1020–990 cm^{-1}) of the 10.0 wt% urea cream, a collection of 10 different spectra, using the WAI scheme. The 1005–1001 cm^{-1} range is also enlarged in the figure. All spectra are nearly identical, without significant band shifts. As the illumination area of the WAI scheme is much larger than that of a conventional scheme, consistent spectral features of the urea cream were obtained, although different areas of the sample were illuminated. The band positions and shapes from the averaged spectrum [gray line spectrum in Fig. 4(a)] are almost identical to those obtained using the WAI scheme.

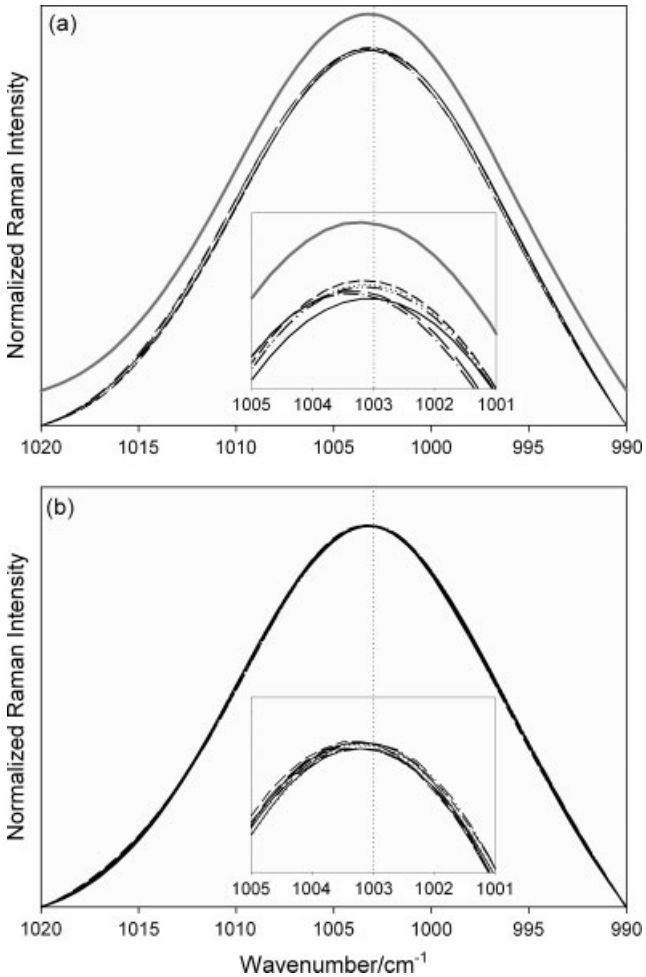

Figure 4. (a) Six different average spectra acquired from each mapping in the 1020–990 cm^{-1} range. The 1005–1001 cm^{-1} range is enlarged in the figure. The spectrum designated with a gray line is obtained by averaging all of the spectra from six mappings (total of 3600 spectra) and is slightly offset for clarity. (b) The spectra of a 10.0 wt% urea cream collected 10 times using the WAI scheme are shown. The 1005–1001 cm^{-1} range is enlarged in the figure.

Based on this observation, it would be difficult to provide a specific guideline of minimal sample coverage area for the analysis of a urea cream; however, a fairly large area of at least 750 × 750 μm (nearly equivalent to 600 × 900 μm) should be sampled. The required sample area would also vary depending on the characteristics of the cream sample such as viscosity and

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dispersion. Although the sample coverage area of the WAI scheme is much larger than the required minimal coverage area stated above, there would be no drawback to covering a wider sample area for reliable quantitative analysis. Using the WAI scheme, representative Raman spectra of a viscous cream sample can be easily collected without the justification for a minimal sample coverage area. For further spectral collection, the WAI scheme was solely used.

Raman spectral features of urea cream in a plastic bottle

Pharmaceutical products are frequently stored in plastic containers. A method that enables nondestructive and direct measurement through a container would be great practical interest to those responsible for production and quality assurance. To add analytical value to the proposed Raman measurement, we attempted to collect Raman spectra of samples directly through plastic bottles (barriers). Figure 1(b) shows the uncorrected Raman spectrum of 10.0 wt% urea cream in a plastic bottle and that of the empty plastic bottle. The Raman spectrum of the empty plastic bottle shows the characteristic spectral features of LDPE, such as CH_2 bending at 1454 cm^{-1} and CH_2 twisting at 1295 cm^{-1}. The bands of the plastic bottle in the 1545–1050 cm^{-1} range significantly overlap with the bands of the components in the urea cream, because the components contain a high proportion of hydrocarbon structure; on the other hand, the urea band at 1003 cm^{-1} is clearly observable, without serious overlap with bands of the plastic bottle.

Normalized Raman spectra of urea creams in three different concentrations collected in plastic bottles are shown in Fig. 1(c). The normalization procedure was applied as described above. The urea band in the 1030–980 cm^{-1} range is also enlarged. The overall spectral features were quite similar to those obtained using a quartz cuvette, as shown in Fig. 1(a), except for the addition of spectral features from the plastic bottle. The urea band at 1003 cm^{-1} increased according to the urea concentration. Based on examination of the triplicate spectra from each sample, the urea band is less reproducible when a sample in a plastic bottle is directly measured. In the case of spectral collection using the quartz cuvette, the triplicate spectra are visually almost identical, while there is more variance in the direct bottle measurements. The plastic bottles were rotated three times for the triplicate measurements, with the illuminated areas having different morphologies, optical transparencies and thicknesses. Therefore, less reproducibility was expected, which would decrease the accuracy of the urea concentration determination. Thus, the utility of the direct bottle measurement depends on whether the trade-off of accuracy for simplicity is acceptable in a given situation.

Quantitative determination of urea concentration

Although the urea band at 1003 cm^{-1} is not totally baseline-separated, as shown in Fig. 1, it is quite well isolated from the bands of the other cream components. For the determination of urea concentration, a univariate calibration method using the peak under the urea band at 1003 cm^{-1} was initially attempted. First, the raw Raman spectra were baseline-corrected and normalized as described in Fig. 1, and then the peak area in the 1030–980 cm^{-1} range was calculated. A total of 15 samples were divided into 10 samples for the calibration set and 5 samples for the validation set. Regression using the peak area was performed using both spectral datasets collected using a quartz cell and a plastic container.

Table 1. Results of univariate regression from both spectral datasets collected either from a quartz cell or directly through a plastic container

Data collection method	SEC (wt%)	SEP (wt%)
Spectral collection using a quartz cell	0.11	0.10
Spectral collection directly through a plastic bottle	0.52	0.51

Table 2. PLS prediction results in two different spectral ranges

Spectral range (cm^{-1})	SECV (wt%)	SEP (wt%)	MPE (%)
1545–755	0.10 (6)	0.13	1.06
1030–755	0.09 (4)	0.11	0.92

The numbers in parentheses correspond to the number of factors used.

The results are summarized in Table 1. Both the standard error of calibration (SEC) and the standard error of prediction (SEP) were significantly increased when the Raman spectra were collected through a plastic container. This increase in accuracy was expected because the urea bands at 1003 cm^{-1} are much less reproducible (refer to Fig. 1(c)) when the sample was measured directly in a plastic bottle. In contrast, an accurate determination of urea concentration was achievable with an SEP of 0.10 wt% when the Raman spectra were collected from a sample transferred into a quartz cell.

Although the urea band is fairly selective even within the cream matrix, spectral collection through a plastic barrier substantially influenced the reproducibility of the urea band. To improve the accuracy of the measurement through a plastic container, multivariate calibration methods were used to handle the uncontrollable spectral variations derived from differences in plastic morphology, optical transparency and/or thickness. In this study, PLS regression[21,22] was employed to determine the concentration of urea in a cream matrix contained in a plastic container. Normalized spectra, as shown in Fig. 1, were used for the PLS. We have developed PLS models by cross-validating all samples in the whole dataset (15 samples, total of 45 spectra). Initially, we fixed the number of samples for the calibration and validation sets to 10 and 5, respectively. A large number of combinations were possible in determining which five samples would be assigned to the validation set ($_{15}C_5 = 3003$ cases). By examining the errors for all possible combinations, the standard error of cross-validation (SECV) was calculated.

Along with the use of the full spectral range (1545–755 cm^{-1}), we have developed PLS models using the 1030–755 cm^{-1} range, excluding the 1545–1030 cm^{-1} range where the major bands from the plastic bottles and the base cream components were dominant. The PLS results are shown in Table 2, with the numbers in parentheses corresponding to the number of factors used. Clearly, when the 1030–755 cm^{-1} range was used for the PLS, fewer factors were used. By excluding the spectral region that contains high-intensity bands from the plastic bottle and base cream, the variation from a change of urea concentration can be highlighted with the use of four factors.

To validate the calibration models, Raman spectra of the same urea creams (15 samples, prepared separately) were independently collected at two different periods, and the resulting spectra

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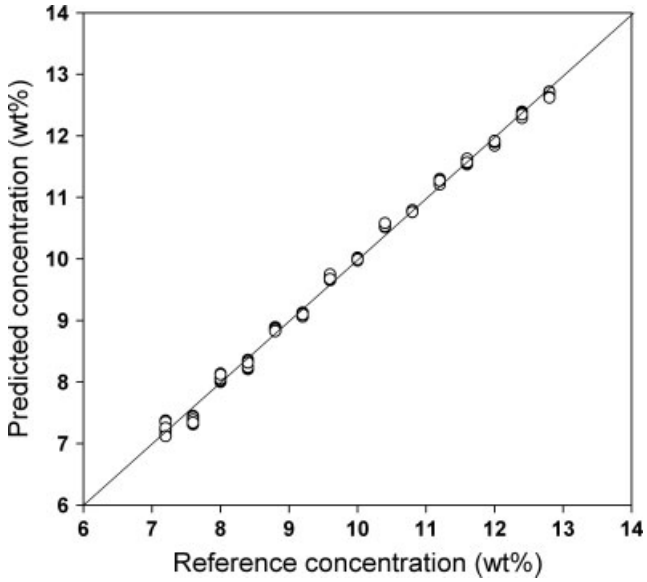

Figure 5. Prediction plot showing the correlation between the reference and predicted values using the cross-validated four-factor PLS model developed in the 1030–755 cm^{-1} range.

were predicted using the models in Table 2. As described in the experimental section, spectra were collected in triplicate for each sample. A total of 90 spectra (45 spectra per dataset) were predicted. As shown in the table, the cross-validated four-factor PLS model using the 1030–755 cm^{-1} range provided more accurate prediction, with an SEP of 0.11 wt% and a mean percent error (MPE) of 0.92%. It is evident that inclusion of the 1545–1030 cm^{-1} range adversely affects the PLS calibration. The SEP was almost identical to that obtained from the experiment using a quartz cell for spectral collection. The MPE obtained in this study was comparable with that of the colorimetric method,[17] which requires an error of less than 1.0%. The prediction plot showing the correlation between the reference and the predicted values is shown in Fig. 5. Although the spectra were separately collected at two different periods, there was no indication of degradation in prediction accuracy.

Conclusions

The importance of correct sample representation for the analysis of cream samples has been critically demonstrated when Raman spectroscopy is used. As long as the features of the entire sample, instead of localized information, can be reproducibly obtained, Raman spectroscopy is a fast, simple and nondestructive analytical method for highly viscous emulsion samples, such as pharmaceutical and cosmetic creams. The WAI scheme is an effective method that ensures representative sampling without the need for sample stirring or rotation during spectral collection, enabling samples to be measured directly through plastic bottles. This method has great potential as a robust and practical Raman spectral collection scheme for the analysis of diverse cream formulations as well as other highly viscous samples.

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